

Karthik Paila

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Objective

Computer Science undergraduate (CGPA 9.26) with hands-on experience in full-stack and AI-powered application development, including RAG pipelines and LLM integrations. Top 5% globally on LeetCode (1850+ rating) across React/Next.js, Node.js/NestJS, Flask, and MongoDB/PostgreSQL. Passionate about building scalable software and exploring practical AI applications.

Education

Rajiv Gandhi University of Knowledge Technologies

Srikakulam, Andhra Pradesh, India

B.Tech – Computer Science **CGPA: 9.26**

Aug 2023 – Apr 2027

Pre University Course (MBiPC) **CGPA: 9.59**

Dec 2021 – Aug 2023

Technical Skills

Languages Python, JavaScript, TypeScript, SQL

Frontend Next Js, React, React Native Expo, Redux, Tailwind, Framer Motion

Backend Node.js, Express, NestJS, FastAPI, RESTful APIs, GraphQL (Basics), WebSockets

AI/ML LLMs, LangChain & LangGraph, RAG, Embeddings, Vector DB (pgvector)

Databases MongoDB, PostgreSQL, MySQL, Redis, BullMQ, Firebase

Tools & Devops Git, Docker, AWS (Basics), Figma

Coding Achievements

- Participated in **4 hackathons**; achieved 1st Runner-Up at Hack24 (MVGR College) for **Campus Matrix**, an interactive campus mapping and navigation system.
- Solved **1500+** DSA problems on **LeetCode** & **GFG**; maintained an **1850+** LeetCode rating (**Top 5%, Knight Badge**).
- Selected as the **#1** Campus Startup for **Apteez**, a competitive aptitude platform recognized for its retention & market potential.
- Completed **50+** advanced SQL challenges on LeetCode and maintained a **98%** solution success rate.
- NPTEL** — Artificial Intelligence: Search Methods for Problem Solving **82% (Silver Badge)**

Projects

ApteeZ | Competitive Aptitude Platform

[🐙 Repo](#) [🔗 Figma](#)

Next.js · TypeScript · NestJS · PostgreSQL · Prisma · Redis · Socket.IO · BullMQ · LangChain.js · LangGraph.js · pgvector

- Building a **full-stack competitive aptitude platform** with 1v1 challenges, timed contests, ratings, leaderboards, structured learning, events, discussions, contributions, and performance analytics across multiple aptitude domains.
- Engineering **real-time matchmaking and competitive gameplay** with Redis and Socket.IO, plus **agentic AI and RAG pipelines** using LangChain.js, LangGraph.js, tool calling, embeddings, and pgvector for **performance coaching, semantic problem retrieval, and AI-assisted question review**.

AI Study Companion | AI-Powered Document Learning Platform

[🐙 Repo](#) [🔗 Figma](#)

Next.js · TypeScript · Express · Prisma · PostgreSQL · pgvector · Redis · BullMQ · Groq · Gemini · Neon Object Storage

- Building a **production-oriented AI learning platform** that converts PDF study material into searchable knowledge and provides **grounded tutoring, adaptive assessments, concept mastery, growth analysis, and personalized recommendations**.
- Engineered an **end-to-end document intelligence pipeline** with asynchronous PDF/OCR processing, Gemini embeddings, **hybrid PostgreSQL + pgvector retrieval**, and Groq-powered Tutor and assessment workflows; implemented **citation-validated RAG, adaptive question selection, mastery updates, and BullMQ-based background processing** with persistent learning and AI usage telemetry.

Risk View | Location-Aware Travel Safety Intelligence Platform

[🐙 Repo](#) [🔗 Figma](#)

React Native Expo · Redux · Node.js · Express · FastAPI · MongoDB · RAG · Nominatim · Groq API

- Built a **location-aware travel safety platform** that enables travelers to discover nearby scam and danger reports, submit community incidents, and monitor **real-time safety information through an interactive map**.
- Engineered an **AI-assisted safety intelligence pipeline** for report validation, incident summarization, and **RAG-based retrieval over community reports**, enabling context-aware answers to safety queries while combining geospatial discovery with AI-generated safety insights.

Hackathons (4 participated, 1 top finish)

- Campus Matrix** – Built an interactive campus navigation system with visual map editing, classroom & timetable management, and GPS-based navigation. [🐙 Repo](#)
- X_Auditory** – AI-powered hearing assistance app for environmental sound detection using audio classification, directional and priority-based alerts, speech-to-text, and customizable sound modes. [🐙 Repo](#)
- GramaVikas** – Farmer assistance app for crop disease diagnosis using image classification, government scheme discovery, emergency support, environmental risk alerts, and AI chatbot guidance. [🐙 Repo](#)